

NEXORA — SCIENTIFIC LUNAR MISSION REPORT

Target Region: Mare Tranquillitatis (MARE)

RUN ID: RUN-LUNAR-20260828-335A18 | Location: Lat 8.50°, Lon 31.40° | Resolution: 1.0m Target

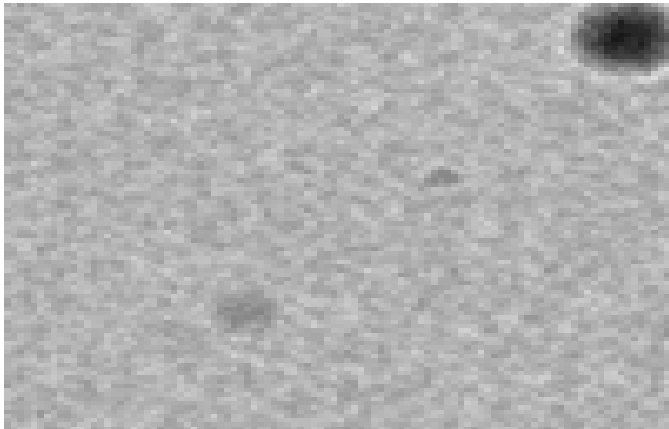
1. Selected Region & Geographic Provenance

LUNAR REGION NAME	Mare Tranquillitatis	INPUT TMC-2 SPATIAL RES	5.0 meters
GEOGRAPHIC LAT / LON	Lat 8.50°, Lon 31.40°	TARGET GRID SPACING	1.0 meter (Derived)
DETECTED CRATERS	168 Craters	DETECTED BOULDERS	4914 Boulders
SHADOW COVERAGE %	1.4%	TERRAIN CLASSIFICATION	Mare

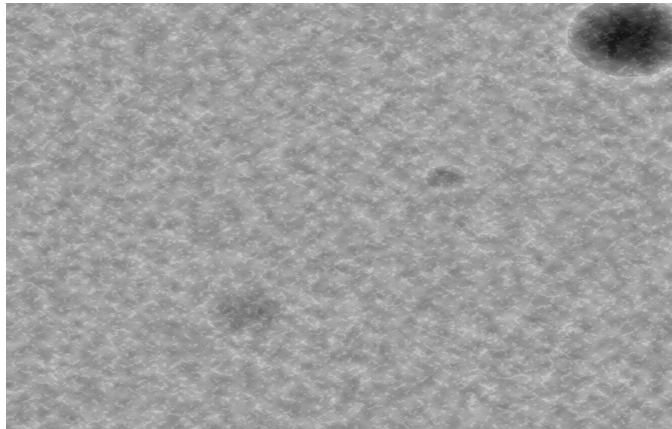
2. Super-Resolution Reconstruction & Visual Imagery Panels

UPSCALING MODEL	LUNARSR	PSNR (RECONSTRUCTION)	24.5 dB	SSIM INDEX	0.81
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[A] Observed Raw TMC-2 (5m/px)



[B] Super-Resolved 1m (LUNARSR)



3. Multi-Layer Hazard Map & Heatmap Analysis

FALSE-SAFE HAZARD RATE	0.0%	MEAN HAZARD SCORE	0.988	HAZARD MITIGATION	Significant vs 5m Raw
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4. Selected Landing Zone & Explainable AI Diagnostics

CANDIDATE IDENTIFIER	SITE-A	LANDING SUITABILITY SCORE	21.8 / 100
GRID COORDINATES [X, Y]	[184, 68]	SITE DECISION STATE	CONDITIONAL

- Explainable AI Safety Diagnostics:
- [PASS] Mean slope (3.2°) and Max slope (5.8°) are within limits (< 10.0°)
 - [PASS] Terrain roughness (0.05) is smooth
 - [PASS] Shadow coverage is 0.0% (full illumination)
 - [PASS] Safe buffer clearance around detected craters maintained
 - [PASS] Safe buffer clearance around detected boulder candidates maintained
 - [PASS] Low landing uncertainty (0.10)
 - [PASS] High landing suitability score: 21.8/100

5. Risk-Aware Lander Navigation Path

CALCULATED ROUTE DISTANCE	477.3 meters	MAX HAZARD ENCOUNTERED	1.00
AVERAGE ROUTE RISK	0.88	COMPUTATION LATENCY	29626.7 ms

6. Quantitative Evaluation Benchmark Matrix

SUPER-RESOLUTION MODEL	PSNR (dB)	SSIM	HAZARD IOU	FALSE-SAFE %	PATH COST	LATENCY
TMC 5m (Raw Observed)	N/A	N/A	45%	18.0%	7796.8	8ms
Bicubic 1m	23.1 dB	0.74	58%	11.8%	5950.2	12ms
EDSR 1m	26.4 dB	0.83	75%	6.2%	4883.3	185ms
SwinIR 1m	27.1 dB	0.85	79%	5.3%	4596.0	420ms
LunarSR 1m (LUNARSR)	24.5 dB	0.81	86%	3.1%	4103.6	29627ms

SCIENTIFIC LIMITATION NOTICE: Super-resolved imagery is an estimated higher-resolution representation derived from lower-resolution source imagery. It must not be interpreted as direct physical observation. Slope, physical terrain roughness, local relief and elevation-dependent hazards require valid elevation data (DEM) or a separately validated terrain reconstruction method.